Voxtral Mini bn-BD

Bengali speech-to-text: how a private fine-tune compares with four public models on seven test sets

Voxtral Mini is the most accurate of the five models on spontaneous and conversational speech (Vaani, SPRING-INX R1), level with the best on FLEURS, and behind Conformer Large and Whisper Medium on read Kathbath speech.

22.1%
word error rate on Vaani
lowest of all five models

36.9%
word error rate on SPRING-INX R1
lowest of all five models

14.7%
word error rate on FLEURS
best public model: 14.4%, too close to call

0.44 s
per 13-second clip on an A100
4× faster than Whisper Medium

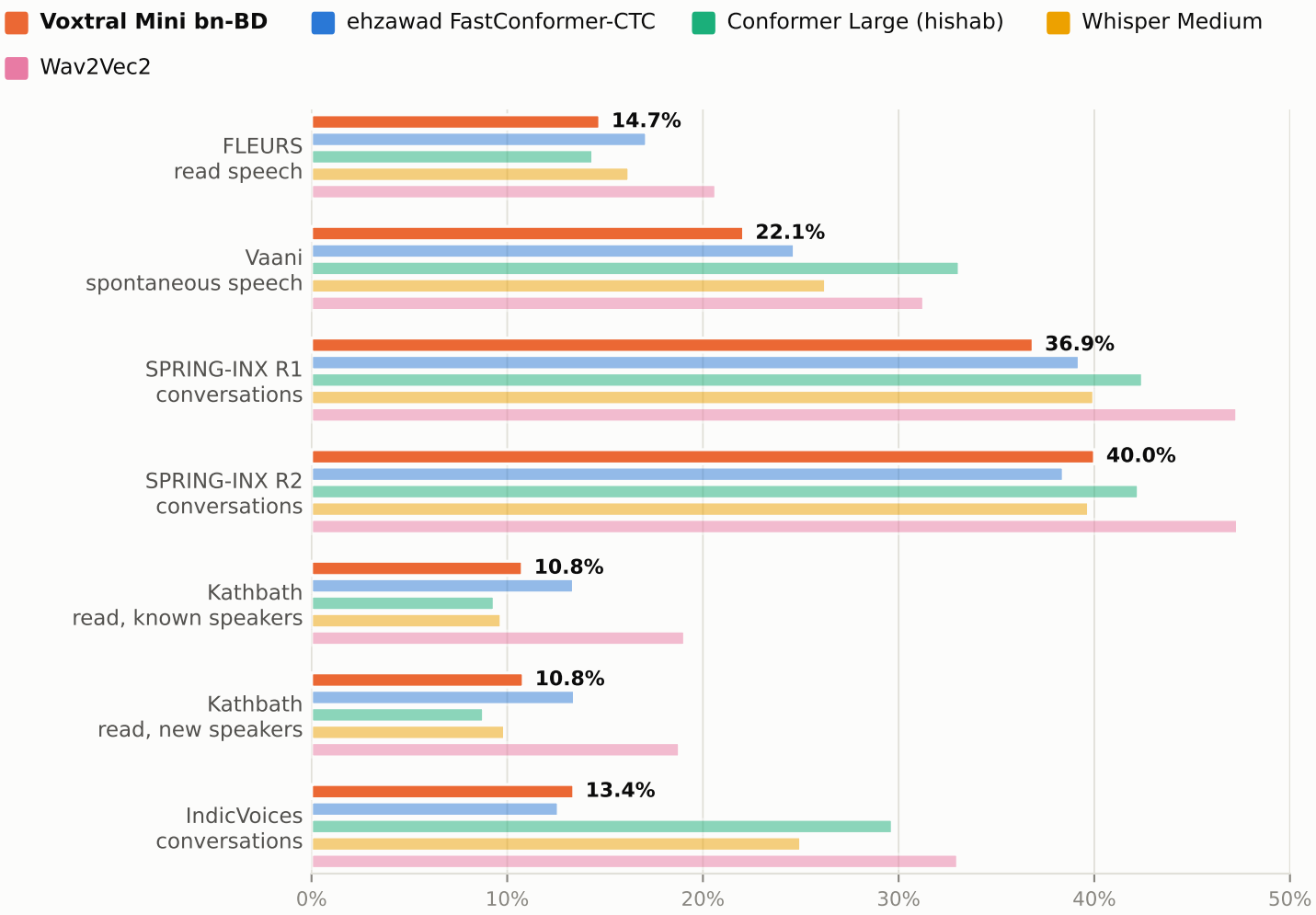
Against the best public model on each test set

Test set	Voxtral Mini	Best public model	Gap	Verdict
FLEURS read speech	**14.7%**	14.4% Conformer Large (hishab)	**+0.4**	≈ Too close to call
Vaani spontaneous speech	**22.1%**	24.7% ehzawad FastConformer-CTC	**−2.6**	✓ Better
SPRING-INX R1 conversations	**36.9%**	39.2% ehzawad FastConformer-CTC	**−2.4**	✓ Better
SPRING-INX R2 conversations	**40.0%**	38.4% ehzawad FastConformer-CTC	**+1.6**	≈ Too close to call
Kathbath read, known speakers	**10.8%**	9.3% Conformer Large (hishab)	**+1.4**	✗ Worse
Kathbath read, new speakers	**10.8%**	8.8% Conformer Large (hishab)	**+2.0**	✗ Worse
IndicVoices conversations	**13.4%**	12.6% ehzawad FastConformer-CTC	**+0.8**	≈ Too close to call

Word error rate (WER): the share of words transcribed wrongly, lower is better. Gap: Voxtral Mini minus the best public model, in percentage points. Better or Worse means the gap holds up in a paired statistical test on the same recordings (95% bootstrap); Too close to call means it does not.

Every model, every test set

Word error rate, lower is better · Voxtral Mini in orange



The numbers

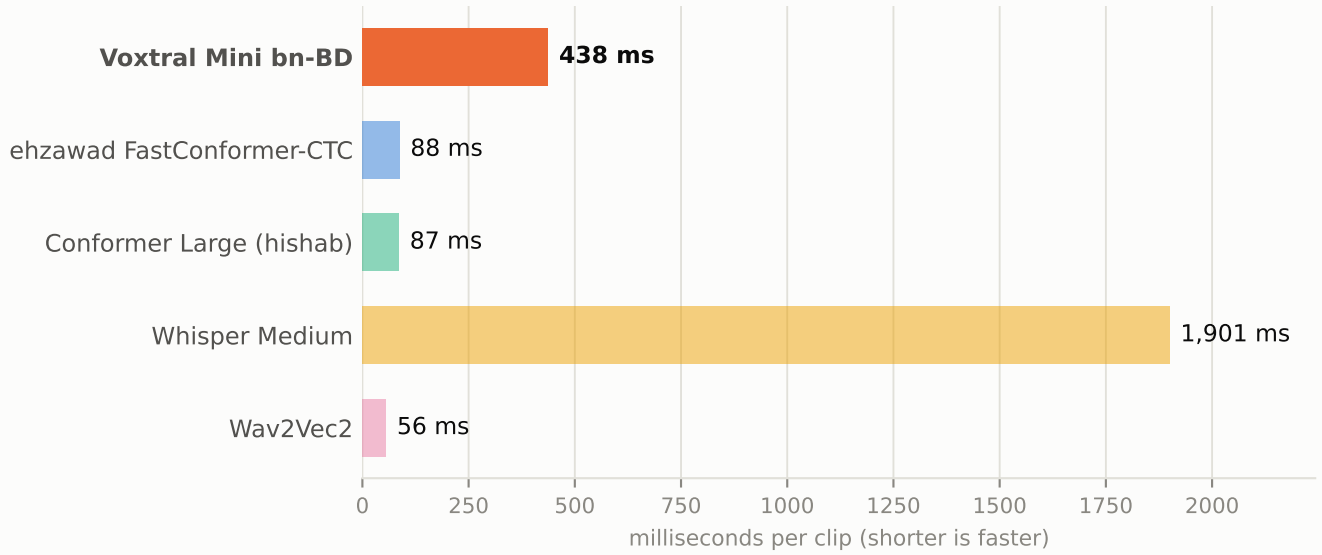
	FLEURS	Vaani	SPRING R1	SPRING R2	Kathbath known	Kathbath new	Indic Voices
Voxtral Mini bn-BD	**14.7**	**22.1**	**36.9**	**40.0**	**10.8**	**10.8**	**13.4**
ehzawad FastConformer-CTC	17.1	24.7	39.2	**38.4**	13.4	13.4	**12.6**
Conformer Large (hishab)	**14.4**	33.1	42.5	42.2	**9.3**	**8.8**	29.7
Whisper Medium	16.2	26.3	40.0	39.7	9.7	9.9	25.0
Wav2Vec2	20.7	31.3	47.3	47.3	19.1	18.8	33.0

lowest word error rate on that set

- Kathbath and IndicVoices are not clean head-to-heads: ehzawad FastConformer-CTC trained on other splits of both (hence its lead on IndicVoices), and Conformer Large on Kathbath.
- Voxtral Mini's training data is not documented, so overlap with any of these sets cannot be ruled out.

Speed, method and caveats

Average time to transcribe one FLEURS test clip (13.4 s of audio), one clip at a time on one A100



Voxtral Mini takes 0.44 s for a 13-second clip, about 31× faster than real time. That is 4× faster than Whisper Medium and 5× slower than the two NeMo CTC models (FastConformer-CTC and Conformer Large). The times cover each model's whole pipeline, audio loading included, on vLLM for Voxtral Mini and PyTorch for the others, so they compare setups, not just the models.

How it was measured

- One Colab A100 (40 GB) on 13 Oct 2026, one model on the GPU at a time.
- The same audio for every model: 16 kHz, cut at pauses into pieces of at most 25 s.
- Plain decoding: greedy, one clip at a time, no external language model.
- FLEURS: the 920 test recordings. Other sets: 1,000 recordings each, sampled from the official splits.
- One scorer for everyone: punctuation removed and Bengali text Unicode-normalised before counting errors.

Keep in mind

- These are Indian Bengali test sets; none is a dedicated test of Bangladeshi speech.
- About half of SPRING-INX's reference sentences contain English words written in Latin letters, which no model produces, so SPRING error rates are inflated for every model alike.
- On SPRING-INX R2, Voxtral Mini got stuck repeating one word on 3 recordings; those alone account for 663 of its 8,143 word errors there, and they stay in its score.
- Like Conformer Large and Whisper Medium, Voxtral Mini writes numbers as Bengali words, which counts as an error wherever the reference uses digits.